

Hyperloop Development Program

Reporting Period 2: Jul 2021-Dec 2021

28-02-2022 – V1

This report highlights the progress of the Hyperloop Development Program. First section summarizes the highlights within the growing hyperloop market, the second section provides an overall program overview, followed by section of progress per working group, and summarized in the financial overview.

HIGHLIGHTS WITHIN HYPERLOOP MARKET

The past months have shown highlights in the Global developments in the Hyperloop Market:

- In **August** the US infrastructure bill included provisions to Hyperloop projects, making companies eligible for federal funding for hyperloop projects - [Link](#)
- In **September** the HyperMotion Frankfurt was held. The first time all hyperloop stakeholders and major industry players will come together to discuss developments, challenges and solutions, to showcase and to jointly work on main issues in the many workshops - [Link](#)
- In **September** Virgin Hyperloop revealed that they will be showcasing a full-scale hyperloop cargo pod and a cutaway passenger pod as part of DP World's FLOW pavilion at Expo 2020. - [Link](#)
- In **October** the seven largest players in the hyperloop industry (Hardt, HyperloopTT, Nevomo, Swisspod Technologies, TransPod, Virgin Hyperloop and Zeleros) and the European Rail Infrastructure Managers (EIM), signed an agreement supporting integration of hyperloop and rail systems.- [Link](#)
- In **October** Hardt Hyperloop has been awarded 15 million euros from Brussels's EIC Accelerator for hyperloop development - [Link](#)
- In **October** Transpod started with the construction of hyperloop test track in Limoges, France - [Link](#)
- In **October** Zeleros published its perspective on a global hyperloop network - [Link](#)
- In **December** it became clear that HTT is scaling down its efforts in Toulouse Test Facility - [Link](#)
- In **December** hyperloop has been included in Proposal TEN-T Regulation. Article 44 states "*In order for the trans-European transport network to keep up with innovative technological developments and deployments, Member States shall aim in particular to: support and promote the decarbonisation of transport [...] such as hyperloop.*" - [Link](#)
- In **December** hyperloop and maglev have been included in the latest 5 year plan of the Chinese Ministry of Rail - [Link](#)
- In **December** the Indian government started a hyperloop feasibility study to explore its technological maturity, financial and commercial viability, safety standards and regulation; to be executed by the Indian think tank / planning commission Niti Aayog - [Link](#)

HDP published 4 communication moments based on deliverables of the Program in the past half year:

- Cargo-Holland (October 2021): 22 mentions in public articles [Link](#)
- Outlook NL (December 2021): [Link](#)
- Research infrastructure map (December 2021): 1 mention in public article [Link](#)
- Certification Roadmap (December 2021): 1 mention in public article [Link](#)

HDP has focused on three means to raise additional funding:

- *Nationaal Groeifonds*: The Stichting HDP started to search for a director to lead the application for Growth Fund round 3 (Q2-Q3 2022). The Director will have as core task to secure future funding through the Groeifonds, and will lead the process.

- *NWA*: in the context of connecting Dutch science, support was given to a project proposal for scientific research on the hyperloop in the field of social value and technology, led by TU Delft in collaboration with more than 10 Dutch research institutions, for the NWA. NWO is evaluating this and the outcome is expected in March 2022.
- *Interreg*: The application for Interreg North-Sea Regions subsidy has been set on-hold until further notice, expected to be picked up in Q3-2022.

HDP has onboarded in total five partners to the program, thus mobilizing more than €1.3M additional funding to the Program.

- Participation of partner IHC will be scaled down, and will remain as an Ecosystem Partner involved. The financial plan for 2022 and 2023 will reflect the commitments into eligible costs for subsidy in the Program.

Partner	Type	Working Group / WP	Commitment
DENYS	Core Program Partner	WG Cargo, WG EHC	1 million EURO
UNStudio	Work Package Partner	WG Cargo, Passenger	100.000 EURO
MERCON	Work Package Partner	WG Cargo	150.000 EURO
AndAnotherDay	Ecosystem Partner	WG Future Prospects	40.000 EURO ¹
Dutch Boosting Group	Ecosystem Partner	n/a	10.000 EURO ²
IHC	Ecosystem Partner	n/a	n/a

¹ will not make any eligible cost for the subsidy

² will not make any eligible cost for the subsidy

REPORTING ON OVERALL PROGRAM PROGRESS

This report is to confirm costs incurred and progress in the Hyperloop Development Program over the period Jul 2021-Dec 2021. The key program deliverable and key deliverable for this stage per working group are represented below.

WG	Key Program deliverable	Key deliverable Stage 2
WG Cargo	Demonstrated hyperloop for cargo system (TRL-7) in the EHC	Preliminary Design Finished
WG EHC	European Hyperloop Center (EHC) Phase A completed	Basic Design Finished, Critical funding secured, Partners/suppliers contracted for Manufacturing & Assembly, Permits approved
WG FP	Identify and develop 5 routes up to a preliminary level, of which 1 with a go-ahead build-out of route	Pre-feasibility study Cargo-Hyperloop Holland finished
WG Passengers	Passenger hyperloop system concept (TRL-3 & SRL-5)	Business Opportunity Definition finished
WG Program Management	Establish a connected hyperloop ecosystem	Growth Fund round 3 application started

As the EHC Working Group experienced some delay in permit application, the partners have not yet been contracted for the construction. As a result, the Consortium Agreement is on-hold until clarity on the EHC Working Group. The overall HDP program expenditure is delayed compared to planned, due to delay within WG EHC, which can be seen in the financial section of this report.

HDP will be finalizing Stage 2 (July 20 – March 22) of a total of four stages. The progress of the Working Groups and Work Packages is presented in figure 1. Green means the Work Package is delivered, Yellow the Work Package is on-going.

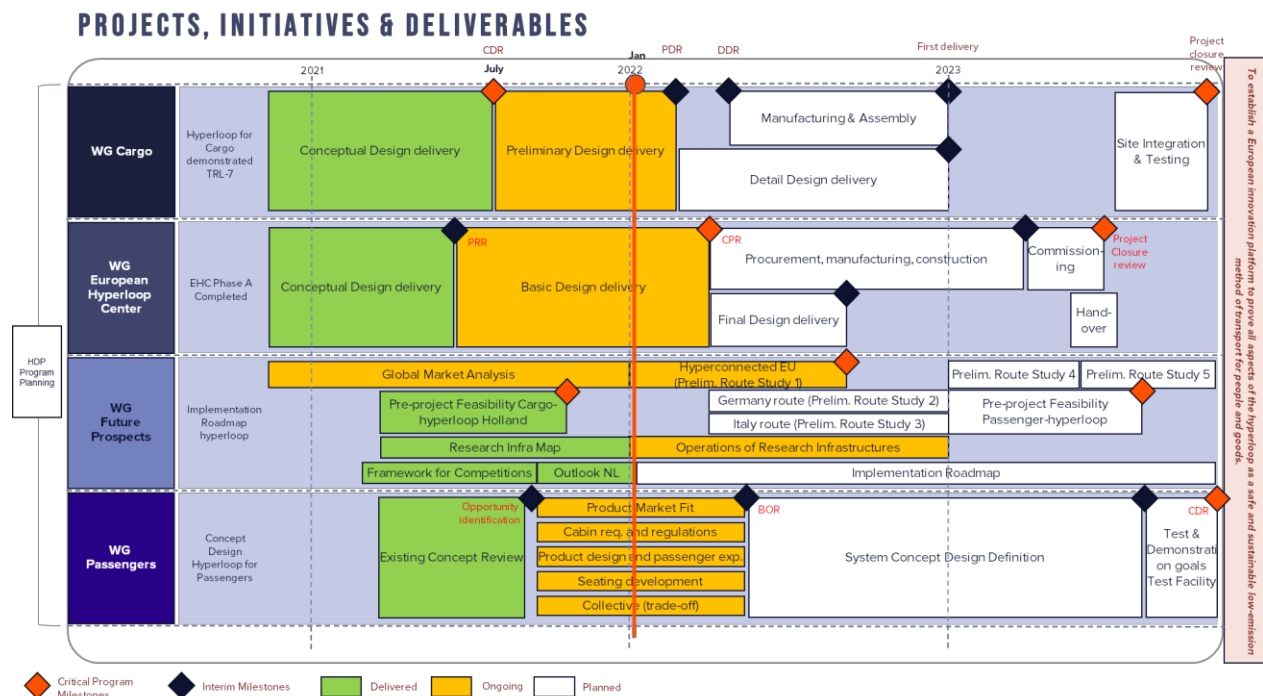


Figure 1: Stages & Milestone planning HDP

The detailed overview of program deliverable status for stage 2 is presented below. A number of milestones from stage 1 have encountered delays, and will be completed in first or second quarter of 2022.

#	Milestone	Working Group	Status	Date delivery
Open from Stage 1:				
1.7	EHC Permits applied	EHC	Delayed	Q2-22
1.9	Final Market opportunity attractiveness	Future Prospects	Delayed	Q1 22
1.16	Consortium Agreement HDP signed	Program Management	Delayed	Q1 22
For Stage 2: Jul 21 - Mar 22				
2.1	Basic Design EHC Approved	EHC	Changed	01/04/2022
2.2	Critical Funding and resources are secured for EHC.	Program Management + EHC	Changed	01/04/2022
2.3	EHC Permits approved (Basic Design Approved)	EHC	Delayed	01/04/2022
2.4	Partners/suppliers contracted for Manufacturing & Assembly.	EHC	Changed	01/04/2022
2.5	Preliminary Design Cargo approved	Cargo	On-track	01/02/2022
2.6	Pre-project feasibility Cargo finished	Future Prospects	Finished	01/09/2021
2.7	Business Opportunity Definition completed	Passengers	On-track	31/03/2022
2.8	Scheme of operations for Hyperloop Research Infrastructure	Future Prospects	Future	31/09/2022
2.9	Check status partners & Stakeholders Green light	Program Management	On-track	31/03/2022
2.10	Plans approved and funding secured for stage 3	Program Management	Future	31/03/2022
2.11	Interreg NSR subsidy secured	Program Management	Cancelled	31/03/2022
2.12	NWA ORC Subsidy secured	Program Management	Waiting for approval	01/01/2022
2.13	Growth Fund round 3 started	Program Management	Future	01/01/2022

In 2021-H2 four expert sessions have been organized on the research themes of the HDP. The purpose of the sessions is to review work, share implications and insights about the work being done in the HDP, with the aim of knowledge sharing and transfer.

- Technology in July based on the work of Conceptual Design phase of WG Cargo.
- Socio-Cost Benefit / Integration in September based on the work of Global Market Assessment and Pre-project feasibility cargo.
- Ecosystem in September based on the work of Inventory of Testing Facilities, Hyperloop Competitions, and Roadmap NL.

In 2022-H1 six expert sessions will be organized on the research themes of the HDP:

- Technology as review of preliminary design Cargo (February)
- Standards and Safety: on Certification Roadmap (February)
- Public Adoption on passenger Working Group (March)
- Socio-Cost Benefit on Preliminary route study #1 (May)
- Integration on the passenger Working Group (May)
- Ecosystem on work need for Hyperloop Research Infra (April)

REPORTING PER WORKING GROUP

WG Cargo

Status: Proceeding according to WG Plan

The Working Group Cargo finished reporting of the Conceptual Design Phase in July, and progressed to Preliminary Design Phase. Current plan is the WG rounds up the Preliminary Design Phase by January 2022, a slight delay of one month to original planning. From February onwards the Working group will deliver the Detail Design, following plan by Q4 2022. Further, A certification roadmap has been prepared by TUV Rheinland as a guideline to work toward testing and acceptance of hyperloop.

The team has finished two main deliverables:

- [HDP-WG-CARGO-1.1]- WG Cargo - Conceptual Design Phase Summary
- [HDP-WG-CARGO-1.19]-Certification RoadMap for the Hyperloop project of Hyperloop Development Program

Next key milestone:

- [HDP-WG-Cargo-2.5] - Preliminary Design Summary Document Cargo (Jan 2022)
- [HDP-WG-Cargo-3.8] - Detail Design Summary Document Cargo (Q4 2022)

WG Passengers

Status: Proceeding according to Schedule

The Working Group Passenger is working on the findings of the Passenger Concept Review of the first half year 2021. The Working Group works on three design scenarios, Small, Medium, Large, to investigate and trade-off what are the design implications for the passenger when choosing one of the scenarios. The primary focus on the cabin, safety and interior. The Working Groups expects to finish the stage by March 2022.

Interim deliverable status:

- Passenger hyperloop Concept of Operations (CONOPS) – On-going
- Report on requirements and regulations – On-going
- Report on safety vision – On-going
- Product design and passenger experience – On-going
- Seating development – On-going
- Trade-off process – On-going
- Report on the Business Opportunity Definition (BOD) – On-going

Next key milestones:

- [HDP-WG-Pax-2.7] - Delivery Business Opportunity Definition (March 2022)
- Delivery of plan for next phase (March 22)

WG EHC

Status: Change to baseline

Over the course of 2021 the Working Group experienced delays in the preparations for the European Hyperloop Center. With the funding of the European Commission for Hardt, the milestone *critical funding and resources secured for EHC* is on-track for Q1-2022, but that means the legal procedures are the bottleneck for swift delivery of the project.

Therefore, to increase speed, the Working Group is exploring a temporary buildup of the EHC phase A on a different location near the existing site, in Veendam. In time the temporary phase A will be transported to the original location. The Working Group is reviewing the location and the impact on the project, and a definitive decision for this approach will be made by the end of January. This change would mean the Working Group is able to meet its schedule and start construction by Q2-2022, and deliver the project in 2023.

This does not mean that the plans are abandoned for the original location in Groningen. The Working Group will maintain the pace and will start the permit procedures for the full 2,6 km test track in Meerstad, Groningen, by Q2-2022.

Next key milestones:

- [HDP-WG-EHC-2.1] – Basic Design Document EHC Phase A (Q1-2022)
- [HDP-WG-EHC-2.2] – Critical Funding and resources are secured for EHC. (Q1-2022)
- [HDP-WG-EHC-2.3] – EHC Permits approved (Q2-2022)
- [HDP-WG-EHC-2.4] – Partners contracted for Manufacturing & Assembly Phase A (Q1-2022)

WG Future Prospects

Status: Proceeding according to WG Plan

During H2-2021 WG Future Prospects was able to publish several deliverables that provided valuable insights for the HDP partners and the wider hyperloop community. Simultaneously the publication of those research items allowed the HDP to gain broad attention as the research was cited in a multitude of articles of public media outlets. The WG was also able to close some of the gaps from the WG plan, which involved onboarding AndAnotherDay for the task “interactive network tool” of WP1 and Berenschot for task “Roadmap NL” of WP7.

The WG also experiences some delay for certain deliverables, which is being further elaborated in the table below. Furthermore, it was not possible to find partners for all activities that were scheduled up until the end of reporting period H2-2021, which leads to a reassessment of the necessity of certain research items. The outcomes of research already performed has given insights that already led us to come to the decision not to continue with certain task, e.g. the actual execution of hyperloop competitions. Those aspects are also further elaborated in the table below.

Program Milestone ID	WP	Title	Status
HDP-WG-FP-1.9	1	Final Market opportunity attractiveness	Ongoing
HDP-WG-FP-1.10	3	Alignment preliminary market needs/ wishes/ requirements of Pre-project feasibility Cargo	Completed
HDP-WG-FP-1.11	2	Established plan and process to execute the 5 preliminary route studies	Completed
HDP-WG-FP-1.12	7	First outlook for Roadmap NL has been established	Completed
HDP-WG-FP-1.13	6	Framework for a hyperloop competition finished	Completed

HDP-WG-FP-1.18	5	Inventory of Testing facilities completed	Completed
HDP-WG-FP-2.6	3	Pre-project feasibility Cargo finished	Completed
HDP-WG-FP-2.8	5	Report on operations of hyperloop research infrastructure	Not started
HDP-WG-FP-3.4	2	Report on preliminary route study, Nr. 1	Ongoing
HDP-WG-FP-3.5	2	Report on preliminary route study, Nr. 2	Not started
	1	Global and continental network sketches	Completed
	1	Traffic assessment model	Discontinued
	1	Interactive network tool	Ongoing
	2	Progress Report on Local Stakeholder Consultation	Completed
	6	Progress Report on hyperloop competitions	Discontinued
	2	Progress Report on Local Stakeholder Consultation	Not started yet
	5	Report on need for future hyperloop research infrastructure	Not started yet
	5	Report on location selection process for large-scale research infrastructure	Not started yet
		Report on tube and pod material	Ongoing
		Report on energy management	Ongoing
		Report on guideline for hyperloop cost-benefit analysis	Ongoing
	7	Annual hyperloop outlook: the Netherlands	Ongoing
	7	Annual hyperloop outlook: Europe	Not started yet
	7	Annual hyperloop outlook: Global	Not started yet